

Quantitative ^1H Magnetization Transfer Imaging *in Vivo*

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A major factor contributing to proton (^1H) spin-lattice relaxation in biological tissues is believed to be magnetization transfer between ^1H in free bulk water and ^1H in restricted motion associated with macromolecules. We have shown recently that saturation transfer is an effective approach for studying this magnetization transfer process. Herein the determination of magnetization transfer rates in biological tissues is further analyzed by considering the time and power dependencies of saturation transfer. Following these analyses, quantitative magnetization transfer rate constant image maps were collected from the kidney *in vivo*. These rate constant images may prove useful in quantitative tissue characterization and in the determination of tissue-specific ^1H relaxation mechanisms.

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INTRODUCTION

The contrast observed in nuclear magnetic resonance imaging is a complex function of spin density and spin relaxation. In proton (^1H) NMR imaging, the relaxation properties of water ^1H create contrast between different tissue types, both normal and pathologic, which may have the same spin density. The nuclear spin relaxation of water ^1H in biological tissues may involve a number of possible mechanisms and pathways. One mechanism proposed to dominate water ^1H spin-lattice relaxation in biological tissues is magnetization transfer between ^1H in “free” bulk water of relatively unrestricted motion ($^1\text{H}_f$) and ^1H in restricted motion moieties ($^1\text{H}_r$), such as macromolecular protein matrices and cellular membranes (1–6). This process is schematically represented in Fig. 1. Figure 1 also includes the possible contributions from chemical exchange and diffusion. In this model, ^1H in a surface layer of water interacts with the macromolecular matrix via chemical exchange or dipole-dipole interaction. The surface layer of water then communicates with the bulk water pool by diffusion.

Recently, the saturation transfer method (7, 8) has been used to study the magnetization transfer between $^1\text{H}_f$ and $^1\text{H}_r$ and to generate a unique form of NMR image contrast, magnetization transfer contrast (MTC) (6). In this approach, the $^1\text{H}_r$ spins are selectively saturated with radiofrequency (rf) irradiation 5 to 10 kHz off resonance from $^1\text{H}_f$. The 5–10-kHz off-resonance irradiation is effective in saturating the 20–40-kHz broad $^1\text{H}_r$ component with little effect on the narrow (15 Hz) $^1\text{H}_f$ signal (see below). With steady-state $^1\text{H}_r$ saturation, magnetization transfer between the $^1\text{H}_f$ and $^1\text{H}_r$ pools causes a decrease in the $^1\text{H}_f$ NMR signal intensity (i.e., a negative nuclear

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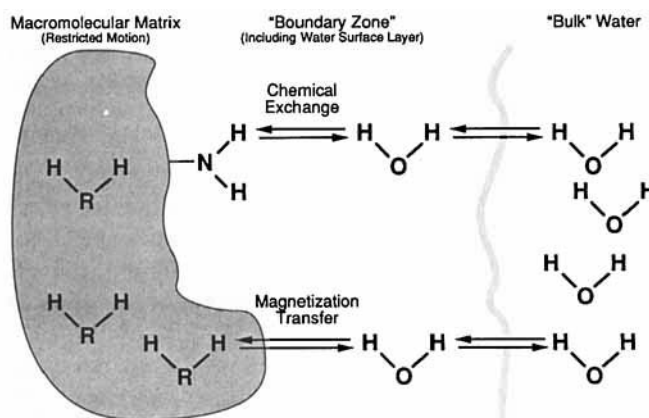


FIG. 1. Schematic diagram of water proton magnetization transfer in biological tissues. Also depicted are the possible roles played by chemical exchange and by diffusion of water to and from the surface layer.

Overhauser effect, NOE) as well as a decrease in the observed $^1\text{H}_f$ spin-lattice (T_1) relaxation time (6, 8). The magnitude of the $^1\text{H}_f$ signal decrease is related to the rate of magnetization transfer by the equation (6–8)

$$k_{\text{for}} = \frac{1}{T_{1\text{fsat}}} \left(1 - \frac{M_{\text{sf}}}{M_{0f}} \right), \quad [1]$$

where k_{for} is the forward pseudo-first-order rate constant of magnetization transfer from $^1\text{H}_f$ to $^1\text{H}_r$ whether the exchange is due to chemical exchange or through-space interactions, $T_{1\text{fsat}}$ is the spin-lattice relaxation time of $^1\text{H}_f$ in the presence of steady-state $^1\text{H}_r$ saturation, M_{sf} is the $^1\text{H}_f$ magnetization in the presence of $^1\text{H}_r$ saturation, and M_{0f} is the $^1\text{H}_f$ magnetization in the absence of $^1\text{H}_r$ saturation. The pseudo-first-order rate constant k_{for} includes magnetization transfer between bulk water $^1\text{H}_f$ and “boundary” water ^1H as well as between boundary water ^1H and macromolecular $^1\text{H}_r$ (see Fig. 1).

Saturation transfer was first described as a method for studying chemically exchanging nuclear species (7, 9), but the analysis is applicable to other mechanisms of magnetization transfer, including through-space dipole-dipole processes (10). Saturation transfer can also be incorporated into NMR imaging pulse sequences in order to map the spatial distribution of the magnetization transfer in intact tissues. On the basis of M_{sf}/M_{0f} ratio images, qualitative imaging of magnetization transfer rates *in vivo* has been used to study the distribution of the creatine kinase reaction in skeletal muscle (11) and of $^1\text{H}_f$ – $^1\text{H}_r$ magnetization transfer in the rabbit kidney (6). The latter study demonstrated that magnetization transfer provides image contrast, termed magnetization transfer contrast, that is not directly related to conventional T_1 or T_2 contrast. However, simple M_{sf}/M_{0f} ratio images do not present the distribution of magnetization transfer rates quantitatively since $T_{1\text{fsat}}$ correction was not applied in these early studies.

The purpose of this study was twofold: (1) to further characterize the saturation power and time dependence of the saturation transfer method for measuring $^1\text{H}_f$ –

$^1\text{H}_r$ magnetization transfer, and (2) to evaluate the quantitative spatial distribution of $^1\text{H}_f$ – $^1\text{H}_r$ magnetization transfer in tissue by obtaining image maps of the transfer rate constant *in vivo*. These analyses should be useful in the development and optimization of imaging protocols incorporating MTC.

MATERIALS AND METHODS

A General Electric 4.7-T CSI system was used for all experiments. ^1H NMR spectroscopy and imaging were performed using a custom-built saddle coil surrounding the surgically exposed rabbit kidney, *in vivo*, as previously described (12). Homogenous agar phantoms (1.5% at 22°C) were used for validating the rate constant image maps using the same coil system.

Selective saturation of the $^1\text{H}_r$ spins was accomplished by applying rf irradiation through the decoupler channel at 5–10 kHz below the $^1\text{H}_f$ resonance. An off-resonance irradiation strength of 1×10^{-5} T was used in these experiments. In selecting an off-resonance irradiation strength (i.e., B_1 field), an attempt was made to optimize $^1\text{H}_r$ selectivity and saturation while minimizing nonspecific radiofrequency “bleedover” effects on $^1\text{H}_f$. The effect of a given B_1 strength on either $^1\text{H}_r$ or $^1\text{H}_f$ is given by the equation (from Ref. (13))

$$m_{\text{ratio}}(u) = \frac{M_{su}}{M_{0u}} = \frac{1 + \Delta\omega^2 T_{2u\text{obs}}^2}{1 + \Delta\omega^2 T_{2u\text{obs}}^2 + \gamma^2 B_1^2 T_{1u\text{obs}} T_{2u\text{obs}}}, \quad [2]$$

where the subscript symbol u is either f or r to denote the $^1\text{H}_f$ or $^1\text{H}_r$ spins, respectively; the function $m_{\text{ratio}}(u)$ is used here to denote the fraction of $^1\text{H}_u$ magnetization remaining along the z axis; M_{su} is the magnetization of $^1\text{H}_u$ observed in the presence of off-resonance irradiation; M_{0u} is the magnetization of $^1\text{H}_u$ in the absence of off-resonance irradiation; $\Delta\omega$ is the off-resonance frequency shift (rad/s); $T_{1u\text{obs}}$ is the *observed* spin–lattice relaxation time of $^1\text{H}_u$; and $T_{2u\text{obs}}$ is the *observed* spin–spin relaxation time of $^1\text{H}_u$. In selecting B_1 , we would like to attain maximal saturation of $^1\text{H}_r$ with minimal effect on $^1\text{H}_f$ (i.e., optimize the difference $m_{\text{ratio}}(f) - m_{\text{ratio}}(r)$). Assuming observed ^1H relaxation times in the kidney of $T_{1f\text{obs}} = 1.5$ s, $T_{2f\text{obs}} = 60$ ms, $T_{1r\text{obs}} = 1$ s, and $T_{2r\text{obs}} = 0.2$ ms (6), the difference $m_{\text{ratio}}(f) - m_{\text{ratio}}(r)$ can be calculated for given values of B_1 and $\Delta\omega$ (Fig. 2). At the B_1 field of 1×10^{-5} T used in these experiments, the calculated nonspecific rf bleedover on $^1\text{H}_f$ was $\sim 15\%$ with only a 5-kHz offset, while $^1\text{H}_r$ was 95–97% saturated. A factor of $m_{\text{ratio}}(f)$ was used to correct for the effect of nonspecific bleedover on M_{sf}/M_{0f} and $T_{1f\text{sat}}$.

The off-resonance B_1 field strength of the saturating field was estimated by measuring the pulse width for a 180° flip of $^1\text{H}_f$, on resonance, through the decoupler channel. The irradiation strength B_1 is given by $\pi/(\tau_{180^\circ} \gamma)$, where τ_{180° is the 180° pulse width and γ is the gyromagnetic ratio given in rad/s/T. This method was validated by measuring the steady-state off-resonance rf bleedover, as a function of frequency, for a ^1H waterline with known T_1 and T_2 . The B_1 field was then calculated from Eq. [2] (14). Both methods provided identical values for the B_1 saturating field.

The method for obtaining quantitative magnetization transfer rate constant maps of the kidney and the agar phantoms involved determining $T_{1f\text{sat}}$, M_{sf} , and M_{0f} for each pixel in the field of view and then applying Eq. [1] to calculate k_{for} on a pixel-by-pixel basis. To determine $T_{1f\text{sat}}$, inversion recovery images were collected at a num-

ber of different inversion times in the presence of $^1\text{H}_r$ saturation. These images were collected with a modified inversion recovery imaging pulse sequence consisting of a variable predelay, a nonselective 180° rf inversion pulse, a variable recovery delay (i.e., the inversion time), and finally a spin-echo imaging sequence to sample the $^1\text{H}_r$ magnetization. A predelay was selected so that the sum of the predelay and the inversion recovery delay was 5 s. A short echo time (18 ms) was chosen for the spin-echo imaging sequence to minimize T_2 contrast. Off-resonance rf irradiation was applied during both the predelay and the inversion recovery delay to achieve $^1\text{H}_r$ saturation.

The $T_{1\text{fsat}}$ image set consisted of five inversion recovery images with inversion times (TI) of 50, 200, 1800, 2400, and 3000 ms. These inversion times were chosen so that each pixel within an image had the same algebraic sign along the inversion recovery curve relative to the other pixels in the same image (i.e., same side of the zero-crossing), assuming a $T_{1\text{fsat}}$ range of 0.3–2.6 s. This choice of inversion times permitted the use of magnitude calculation in the image reconstruction. The TI = 3000 ms image in the presence of $^1\text{H}_r$ saturation was also taken to represent the M_{sf} values. An M_{Of} map (i.e., proton density image) was obtained with TI = 3000 ms without $^1\text{H}_r$ saturation.

Two-dimensional Fourier transform reconstruction of the inversion recovery images, with magnitude calculation, was performed on the CSI system. The reconstructed images were transferred in binary form to a Sun Microsystems 3/260 UNIX-based computer workstation for all other image processing and analysis. The $T_{1\text{fsat}}$ map was calculated by performing nonlinear least squares fits for each pixel location in the $T_{1\text{fsat}}$ image set using a three-parameter inversion recovery equation (15) and corrected for partial saturation. Each least squares fit was performed on five pixel values, one taken from corresponding locations in each image of the $T_{1\text{fsat}}$ image set. Finally, to obtain the k_{for} map, a pixel-by-pixel calculation was performed on the $T_{1\text{fsat}}$, M_{sf} , and M_{Of} maps using Eq. [1].

Software for nonlinear least squares fitting, pixel calculations, and image analysis was developed in the C programming language for a Sun 3/260 workstation equipped with a Pixar image computer for image display. Values are reported $\pm$ SEM.

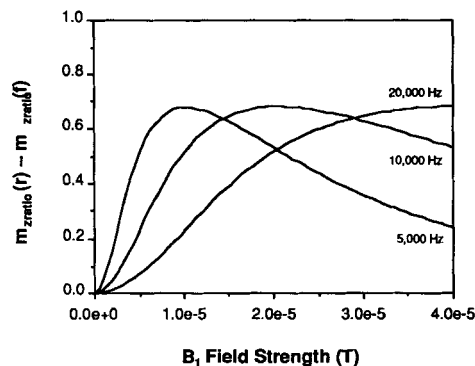


FIG. 2. Plots of $m_{\text{zratio}}(f) - m_{\text{zratio}}(r)$ versus the off-resonance irradiation field strength. The plots are calculated from Eq. [2] for protons using relaxation times ($T_{1\text{fobs}} = 1.5$ s, $T_{2\text{fobs}} = 60$ ms, $T_{1\text{robs}} = 1$ s, and $T_{2\text{robs}} = 0.2$ ms) for the rabbit kidney at 200 MHz (see text). Plots are shown for three off-resonance frequency offsets: 20,000, 10,000, and 5000 Hz.

RESULTS

Power and Time Dependence

The observed M_{sf}/M_{of} ratio is a function of the rate of magnetization transfer (k_{for}) and the relaxation times of both the 1H_f and 1H_r spins. Moreover, in the development of quantitative imaging protocols employing MTC, it is important to evaluate how M_{sf}/M_{of} depends on the duration and strength of the off-resonance B_1 field used to saturate 1H_r , including the effects of nonspecific rf bleedover. The time- and B_1 -dependent development of M_{sf}/M_{of} , assuming instantaneous saturation (complete or partial) of 1H_r at time $t = 0$, can be derived by solving the differential equation

$$\frac{dM_{sf}}{dt} = -\frac{M_{sf}}{\tau_{1f}} + \frac{M_{of}}{T_{1f}} + k_{rev}M_{sr}, \quad [3]$$

where

$$\frac{1}{\tau_{1f}} = k_{for} + \frac{1}{T_{1f}}, \quad [4]$$

T_{1f} is the 1H_f spin-lattice relaxation time in the absence of magnetization transfer, M_{sr} is the 1H_r magnetization in the presence of off-resonance irradiation, k_{rev} is the pseudo-first-order reverse rate constant of the magnetization transfer from 1H_r to 1H_f , and M_{sf} , M_{of} , and k_{for} are defined as above. M_{sr} is considered here to be a constant given by

$$M_{sr} = M_{or} \cdot m_{zratio}(r), \quad [5]$$

where M_{or} is the 1H_r magnetization in the absence of off-resonance irradiation and $m_{zratio}(r)$ is as defined in Eq. [2]. Since the 1H_r linewidth *in vivo* is very broad, M_{sr} and M_{or} are not reliably observable on our spectrometer, and thus all analysis must be made on the 1H_f spins. Note that $M_{sr} = 0$ for the case of complete saturation of the 1H_r spins.

Equation [3] is essentially one equation of the system of differential equations analyzed by Forsén and Hoffman (7, 10) for the chemical exchange experiment. In the steady-state magnetization transfer experiment, however, M_{sr} is a constant (Eq. [5]) so that Forsén and Hoffman's system of equations reduces to a single equation given by Eq. [3]. The solution of Eq. [3] is given by

$$\frac{M_{sf}}{M_{of}} = \left(\frac{\tau_{1f}}{T_{1f}} + \frac{\tau_{1f}k_{rev}M_{sr}}{M_{of}} \right) + \left(k_{for}\tau_{1f} - \frac{\tau_{1f}k_{rev}M_{sr}}{M_{of}} \right) \exp(-t/\tau_{1f}). \quad [6]$$

To see how the steady-state value of M_{sf}/M_{of} varies as a function of B_1 , we let $t \rightarrow \infty$ in Eq. [6] and apply the definition of M_{sr} (Eq. [5]) to obtain

$$\frac{M_{sf}}{M_{of}} = \frac{\tau_{1f}}{T_{1f}} + \frac{\tau_{1f}k_{rev}M_{sr}}{M_{of}} \quad [7]$$

$$= \frac{\tau_{1f}}{T_{1f}} + \frac{\tau_{1f}k_{rev}M_{or}}{M_{of}} m_{zratio}(r). \quad [8]$$

Since the unknowns k_{rev} and M_{0r} are constant and since $M_{\text{sf}}/M_{0f} = 1$ when $B_1 = 0$ (and $m_{\text{zratio}}(r) = 1$), Eq. [8] reduces to

$$\frac{M_{\text{sf}}}{M_{0f}} = \frac{\tau_{1f}}{T_{1f}} + \left(1 - \frac{\tau_{1f}}{T_{1f}}\right) m_{\text{zratio}}(r). \quad [9]$$

To account for nonspecific rf bleedover onto $^1\text{H}_f$ from the off-resonance irradiation, an additional factor of $m_{\text{zratio}}(f)$ must be introduced into Eq. [3], so that its second term, M_{0f}/T_{1f} , is substituted with the term $m_{\text{zratio}}(f) \cdot (M_{0f}/T_{1f})$. This substitution results in a version of Eq. [9] that provides a correction for nonspecific bleedover:

$$\frac{M_{\text{sf}}}{M_{0f}} = \left(\frac{\tau_{1f}}{T_{1f}}\right) m_{\text{zratio}}(f) + \left(1 - \frac{\tau_{1f}}{T_{1f}}\right) m_{\text{zratio}}(r). \quad [10]$$

A single parameter fit of Eq. [10] to experimental data from kidney spectra is shown in Fig. 3. The good correlation between the experimental data and the model suggested by Eq. [10] indicates that the model is appropriate for describing the B_1 field dependence of the magnetization transfer effect.

The time dependence of M_{sf}/M_{0f} in the presence of complete $^1\text{H}_r$ saturation can be demonstrated by setting $M_{\text{sr}} = 0$ in Eq. [6], reducing it to the familiar equation for saturation transfer (see Ref. (7))

$$\frac{M_{\text{sf}}}{M_{0f}} = \frac{\tau_{1f}}{T_{1f}} + k_{\text{for}}\tau_{1f}\exp(-t/\tau_{1f}). \quad [11]$$

Equation [11] demonstrates that upon instantaneous saturation of $^1\text{H}_r$, the $^1\text{H}_f$ magnetization will approach its new steady-state value with a time constant of $1/\tau_{1f}$. This approach was evaluated on experimental data collected from the rabbit kidney. In Fig. 4, the dependence of M_{sf}/M_{0f} on the saturation time in the spectroscopic study of the whole kidney is presented with a least squares fit to Eq. [11]. The value $1/\tau_{1f}$ was determined to be 1.5 s. Assuming a T_{1f} of 2 s, the rate constant k_{for} is calculated (from Eq. [4]) to be 1.0 s^{-1} , in good agreement with previous steady-state measurements (6). The good fit and correlation with steady-state data indicate that this model is appropriate for describing the time dependence of the magnetization transfer process.

Agar Phantom Imaging

To validate the method for obtaining quantitative rate constant maps, k_{for} was determined for a homogeneous agar phantom using both spectroscopic and imaging techniques. Using conventional spectroscopic techniques on the whole sample as previously described (6), k_{for} was 0.34 s^{-1} for the 1.5% agar phantom at 22°C . A rate constant map of the same phantom was constructed from $T_{1\text{fsat}}$, M_{sf} , and M_{0f} data sets, as described under Materials and Methods. The rate constants obtained from these experiments are compared in Table 1, where good agreement between the spectroscopic and imaging methods is seen. These results validate the k_{for} imaging protocol as a quantitative procedure for constructing rate constant maps.

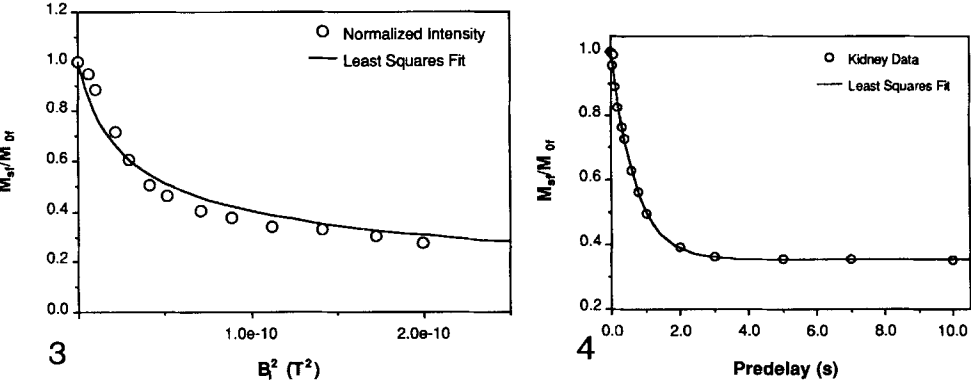


FIG. 3. M_{sf}/M_{of} versus the power of the off-resonance irradiation. The horizontal axis is plotted as B^2_1 , which is proportional to the power of the B_1 field. The data are from a rabbit kidney *in vivo*. The curve is a single-parameter fit of Eq. [10] in which relaxation parameters ($T_{1fobs} = 1.5$ s, $T_{2fobs} = 60$ ms, and $T_{2robs} = 0.2$ ms) for the rabbit kidney are used, as in Fig. 2. The fit revealed a T_{1robs} of 0.14 s.

FIG. 4. Effect of irradiation (saturation) time on M_{sf}/M_{of} in the rabbit kidney. The data were obtained as described in the text. The curve is a two-parameter least squares fit of Eq. [11], revealing a $1/\tau_{1f}$ of 1.5 s.

Kidney Imaging

Following validation, the k_{for} imaging protocol was used to obtain pseudo-first-order rate constant maps of the magnetization transfer process in the rabbit kidney *in vivo*. The proton density (M_{of}) image of the kidney, without 1H_r saturation, is shown in Fig. 5A. A pixel-by-pixel ratio calculation of the 1H_r -saturated, TI = 3000 ms inversion recovery (M_{sf}) image to the M_{of} image yields the M_{sf}/M_{of} image map shown in Fig. 5B. In the M_{sf}/M_{of} image map, 1H_r saturation causes a greater 1H_f signal reduction in the renal cortex than in the medulla. This difference indicates that the magnetization transfer rate is greater in the cortical tissue than in the medulla. The fat and urinary spaces are also much less affected by 1H_r saturation, indicating a low magnetization transfer rate in these compartments. These qualitative observations are consistent with previous magnetization transfer imaging studies of the rabbit kidney (6).

The results of the next step in image processing are shown in Fig. 5C, which is the T_{1fsat} map obtained by performing a pixel-by-pixel inversion recovery curve fit for the five inversion recovery images collected in the presence of 1H_r saturation. The final image-processing step is shown in Fig. 5D, which is a k_{for} map of the kidney obtained by applying Eq. [1] to the corresponding pixels in Figs. 5B and 5C.

The information in the calculated M_{sf}/M_{of} , T_{1fsat} , and k_{for} maps is quantitatively presented in Table 2, which gives average numerical values of each calculated quantity over pixel regions corresponding to the cortex and medulla.

DISCUSSION

As demonstrated in this study, magnetization transfer between 1H_f and 1H_r , as observed by the saturation transfer method, is dependent on both the power and the

TABLE 1
Magnetization Transfer Parameters of 1.5% Agar at 22°C

Method	$T_{1\text{fsat}}$ (s)	$M_{\text{sf}}/M_{\text{of}}$	k_{for} (s^{-1})
Spectroscopic	1.13 ± 0.04	0.62	0.34
Imaging			
Center region (144 pixels)	1.20 ± 0.02	0.58 ± 0.01	0.35 ± 0.01
Whole image (2537 pixels)	1.19 ± 0.04	0.60 ± 0.04	0.34 ± 0.04

duration of the off-resonance rf irradiation. Irradiation powers corresponding to B_1 fields on the order of 1×10^{-5} T at 5–10 kHz away from the $^1\text{H}_f$ resonance can effectively saturate the $^1\text{H}_r$ spins without excessive rf bleedover onto the $^1\text{H}_f$ spins. These low B_1 field strengths are much weaker than those used in spin-locked, off-resonance rotating frame experiments (16), indicating that rotating frame effects are not significant in our experiments. Furthermore, random 90° phase alternation of the off-resonance rf irradiation, which inhibits any spin-locking, does not influence the magnetization transfer effect (unpublished observation).

Even with near instantaneous $^1\text{H}_r$ saturation, the development of the magnetization transfer effect is not instantaneous and depends on both k_{for} and T_{1f} . The experimental data (Fig. 4) and theoretical model (Eq. [11]) show that the $^1\text{H}_f$ and $^1\text{H}_r$ magnetization pools approach equilibrium exponentially with a rate of $1/\tau_{1f}$. An exponential approach, with its steep initial slope, implies that a significant magnetization transfer effect should still be observed in images collected with repetition times (TR) shorter than $1/\tau_{1f}$.

The comparison of k_{for} values obtained by both spectroscopic and imaging methods on agar phantoms demonstrates that quantitative k_{for} maps can be constructed from appropriately collected images. The k_{for} map of the kidney (Fig. 5D) indicates that $^1\text{H}_f$ in the inner medulla has a significantly slower interaction with macromolecular $^1\text{H}_r$ than in the cortex. The k_{for} map of the kidney is consistent with previous $M_{\text{sf}}/M_{\text{of}}$ images of this tissue. However, as seen in Table 2, the magnitude of the difference between cortex and medulla is significantly greater in terms of k_{for} (3.8:1) than in terms of $M_{\text{sf}}/M_{\text{of}}$ (1.6:1). This observation demonstrates that correction for differences in $T_{1\text{fsat}}$ are important and that k_{for} images have significantly greater contrast than in simple M_{sf} or $M_{\text{sf}}/M_{\text{of}}$ images in this tissue. The added contrast comes at a considerable cost in imaging time, however, due to the necessity of determining a $T_{1\text{fsat}}$ image. Faster methods for determining $T_{1\text{fsat}}$ images, such as fast inversion recovery, saturation recovery, or simply acquiring fewer points, may minimize this liability. Although costly in time, the rate constant map protocol is required for quantitative studies of magnetization transfer in biological tissues, such as determining the tissue-specific contribution of magnetization transfer to the proton relaxation process. Rate constant maps may also prove useful in quantitative tissue characterization for diagnostic purposes in magnetic resonance imaging (MRI).

The faster magnetization transfer rate in the cortex as compared to that in the medulla results in observation of MTC in the kidney. MTC also reflects shorter $^1\text{H}_f$

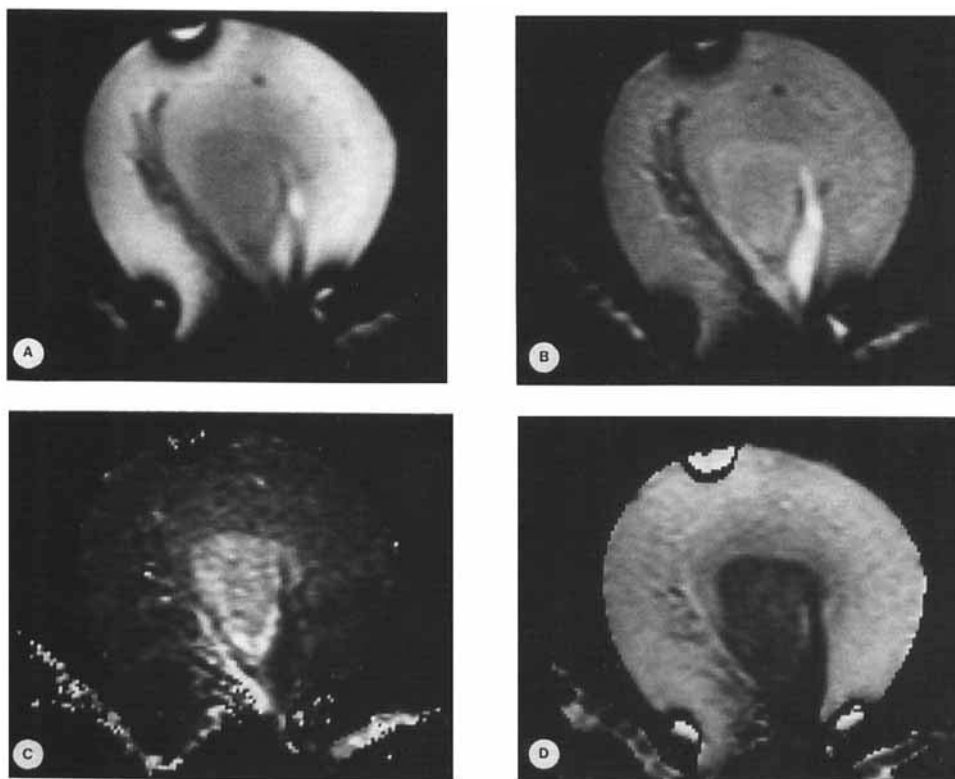


FIG. 5. Images required to construct quantitative rate constant maps of the rabbit kidney *in vivo*: (A) proton density (M_{0r}), (B) M_{at}/M_{0r} , (C) T_{1sat} , and (D) k_{for} images. In all images, the field of view is 60×60 mm with a slice thickness of 3 mm. The proton density image and the T_{1sat} image set were collected with 128 phase-encoding steps followed by zero-filling to a 256×256 digital matrix. The hyperintense spot in the outer cortex of the kidney is due to the close proximity of the coil wire. This region was not used in the analysis of these data.

T_1 and T_2 values in the kidney cortex, as previously discussed (6). The inverse correlation between k_{for} and the relaxation times is consistent with a greater interaction between 1H_f and 1H_r in the cortex of the kidney than in the medulla. The morphological basis for this higher degree of 1H_f – 1H_r interaction remains to be established. Lipid bilayers have been shown to be effective 1H_r macromolecular structures for magnetization transfer with 1H_f (17), and the higher lipid bilayer content of the cortex may be partially responsible for its extensive magnetization transfer effect when compared to that in the medulla.

As stated in the Introduction, the precise mechanism of the observed magnetization transfer effect has yet to be fully described and could involve magnetization transfer via through-space dipole–dipole interactions, chemical exchange, and/or molecular diffusion. If the magnetization transfer process is purely dipole–dipole, then the negative NOE observed in this study corresponds to a rotational correlation time of the water–macromolecule complex on the order of 10^{-8} s. This is a reasonable correlation time

TABLE 2

Magnetization Transfer Parameters of the Kidney *in Vivo*

Kidney tissue	$T_{1\text{sat}}$ (s)	$M_{\text{sf}}/M_{\text{of}}$	k_{for} (s^{-1})
Cortex	0.64 ± 0.08	0.45 ± 0.02	0.91 ± 0.08
Inner medulla	1.3 ± 0.1	0.71 ± 0.02	0.24 ± 0.04

Note. The values under the $M_{\text{sf}}/M_{\text{of}}$ column have been corrected for non-specific rf bleedover by incorporating a factor of $m_{\text{ratio}}(f)$.

for the water-macromolecule complex on the basis of previous studies of protein-water complexes (3, 4).

If chemical exchange is the dominant mechanism, then the effect of $^1\text{H}_r$ relaxation (T_{1r} and T_{2r}) on the observed $^1\text{H}_f$ relaxation times in the presence of magnetization transfer ($T_{1\text{fobs}}$ and $T_{2\text{fobs}}$) is more straightforward, as described previously (6). In this case, $T_{1\text{fobs}}$ and $T_{2\text{fobs}}$ are given by the following equation, assuming no chemical shift difference between the $^1\text{H}_f$ and $^1\text{H}_r$ resonances,

$$\frac{1}{T_{\text{fobs}}} = \frac{1}{T_f} + \frac{X_r}{T_r + (1/k_{\text{rev}})}, \quad [12]$$

where T_{fobs} is the observed T_1 or T_2 of the $^1\text{H}_f$ spins in the presence of magnetization transfer; T_f is the T_1 or T_2 of the $^1\text{H}_f$ spins in the absence of magnetization transfer; X_r is the mole fraction of $^1\text{H}_r$ spins; T_r is the T_1 or T_2 of the $^1\text{H}_r$ spins in the absence of magnetization transfer; and k_{rev} is the rate constant of magnetization transfer from $^1\text{H}_r$ to $^1\text{H}_f$.

Preliminary data suggests that the through-space dipole-dipole mechanism, rather than chemical exchange, is the primary mechanism for magnetization transfer. If water protons are in chemical exchange with a restricted motion pool, then similar effects (i.e., transfer between a broad and free component) should also be observed for other isotopes of protons such as deuterium or tritium. No transfer was observed between broad ^2H spins and water ^2H in tissue *in vivo* (6); but, the relaxation of this particular nuclide could be dominated by quadrupolar relaxation processes and chemical exchange may not be a significant parameter. However, we have also been unable to demonstrate exchange between a broad component of the spin- $\frac{1}{2}$ isotope tritium and water tritium ($\sim 5 \text{ mM}$) in lipid bilayer samples (17) which have typical $^1\text{H}_r$ - $^1\text{H}_f$ magnetization transfer effects (not shown). Thus, the interaction site is apparently nonexchangeable with tritium or deuterium at neutral pH. Furthermore, in model lipid systems and other macromolecules similar magnetization transfer effects have been observed independent of pH from 5 to 8 (6, 17). These results are consistent with the notion that chemical exchange is not a significant factor in the magnetization exchange observed using this saturation transfer technique. Furthermore, the exchange rates of most exchangeable proton sites are either too fast or too slow at physiological pH values to cause an observable magnetization transfer effect (18). It should be pointed out that the diffusional exchange of spins to and from the macromolecular interaction sites is still a potential rate-limiting step in the transfer of magnetization from the

macromolecules to the bulk $^1\text{H}_f$ (5) (see Fig. 1). Diffusion is obviously the rate-limiting aspect of magnetization exchange in dilute macromolecular solutions and could be a significant factor in biological tissues suffering from edema or naturally large free water pools, such as the renal medulla.

In summary, the off-resonance irradiation power and time dependencies for the magnetization transfer experiment have been described along with a theoretical approach for modeling this process. A method for obtaining quantitative pseudo-first-order rate constant maps of this process *in vivo* was presented and validated. Rate constant maps should prove useful in the quantitative characterization of tissue relaxation processes and in the improvement of tissue contrast in proton MRI.

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